

Jitter-capped high-flux single-pixel imaging with global power control

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Abstract

Dead time makes the flux that maximizes an integrated photon count’s information finite, and hidden recovery-time jitter makes that optimum depend on the detector rather than on dwell alone.

For a count-only nonparalyzable detector, an exact missing-information identity shows that uncertainty in the unobserved active time subtracts information from the complete detector path.

In long windows the retained information depends on hold-time variance, and its unique optimum obeys $c_v^2 \rho^2 (1 + 2\rho) = 1$, yielding a jitter cap that scales as $c_v^{-2/3}$.

A single global power multiplier, calibrated from a common pre-scan, implements this detector-specific operating policy without changing the illumination geometry from pattern to pattern.

In a preregistered paired simulation campaign, ridge tracking improved fixed-dwell radiometric quality by a median +1.87 dB, with 19 of 24 scenes positive; this power-for-time comparison used $37.1\times$ the incident dose and $2.6\times$ the detected counts of its matched comparator and is not an equal-photon-budget comparison.

Against the safe-flux reference, the median elapsed-time speedup was $19.13\times$, its family-stratified lower bound was $18.33\times$, and 21 of 24 scenes accelerated.

Separate scene-adapted geometry searches remained descriptive: they exposed feasible headroom under both dose-only and full conditioning constraints but left the scope and reliability of adaptive geometry unresolved.

1 Introduction

Single-pixel imaging (SPI) reconstructs a scene from the responses of one bucket detector to a sequence of structured illumination patterns [?, ?, ?]. When that bucket is a photon-counting detector, every registered event makes it blind for a recovery time. Operating at a few per cent of the dead-time rate avoids strong nonlinearity but spends acquisition time; increasing power gains counts only until dead-time censoring removes more information than the added arrivals provide. The practical question is therefore not simply whether the count response can be corrected, but where a real detector should be operated.

For deterministic recovery, the information-optimal load grows as the cube root of the exposure length [?]. Real recovery times fluctuate. Here we show that hidden hold-time variance produces an extensive information loss, caps the optimal load, and can make the deterministic operating point actively harmful. At $\nu = 2000$, for example, a hold-time coefficient of variation $c_v = 0.05$ moves the optimum from approximately 22.3 to 5.7; using the deterministic ridge then forfeits about 55% of the scalar count information (Fig. 1).

The resulting control is deliberately simple: characterize the detector, maximize its finite-window count-information kernel, and use one global source multiplier to place a fixed balanced pattern family at that load. A preregistered simulation campaign tests the policy at fixed dwell and in elapsed optical time. The policy improves median fixed-dwell quality by 1.87 dB and reduces the time to a frozen quality target by a median $19.13\times$, while explicitly spending $37.1\times$ more incident dose at the terminal dwell. A separate geometry analysis is retained only to locate unresolved headroom; it is not the paper’s positive claim.

Relative to work on high-flux lidar, count-rate correction, and task-specific dead-time optima [?, ?, ?, ?], the contribution is a count-only random-hold law tied to an immediately deployable global power policy and a paired image-level test. The observation is the integrated count, not event timestamps, and all image evidence is simulation-based.

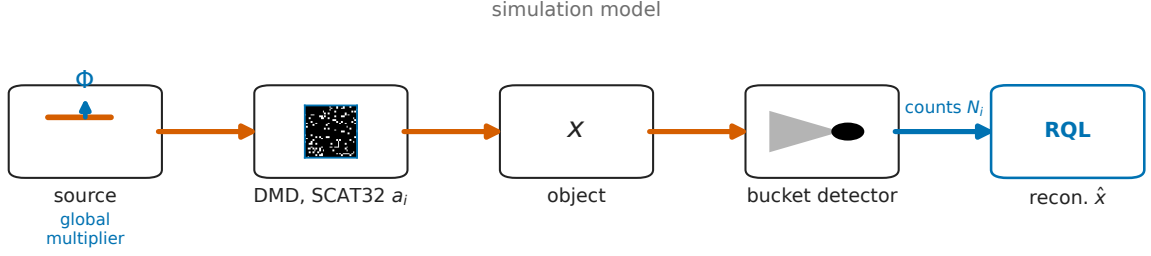
2 Random-hold count information

Let arrivals be Poisson with rate λ , and let every registered event initiate a hidden iid hold $B_i \sim H$, independent of the arrivals and of λ , with mean τ and coefficient of variation $c_v = \sigma_B/\tau$. Only the integrated count N_T over $T = \nu\tau$ is observed; the detector-active time L_T , state path, and realized holds remain hidden. The deterministic model is the special case $c_v = 0$.

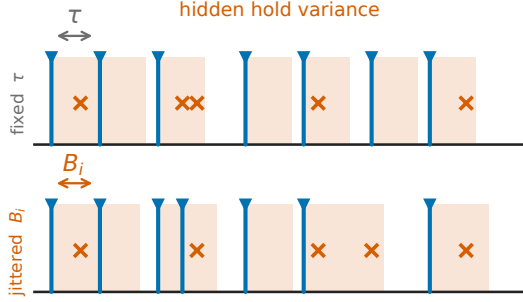
Theorem 1 (Missing-information identity) *For $\theta = \log \lambda$, the complete detector-path information is $\mathbb{E}[N_T]$, and the information retained by the scalar count is exactly*

$$I_N(\theta; T) = \mathbb{E}[N_T] - \lambda^2 \mathbb{E}[\text{Var}(L_T \mid N_T)] \quad (1)$$

(a)



(b)



(c)

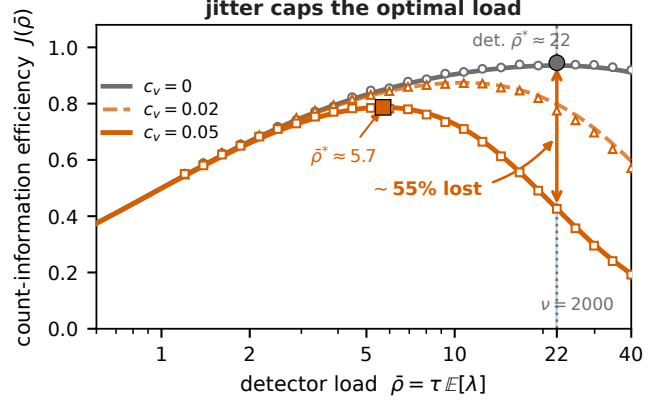


Figure 1: **Hidden dead-time jitter caps the information-optimal flux.** (a) Computational SPI model (simulation model): a global power multiplier Φ scales the source; a DMD displays scattered-sparse SCAT32 patterns a_i on the object x ; a nonparalyzable bucket detector returns counts N_i for renewal quasi-likelihood (RQL) reconstruction. (b) After each registered count the detector is held blind. A fixed hold τ and a random hold B_i censor different arrivals, so hidden hold variance destroys count information. (c) Count-information efficiency $J(\bar{\rho})$ at $\nu = 2000$ for $c_v \in \{0, 0.02, 0.05\}$. The deterministic optimum is $\bar{\rho}^* \approx 22$, whereas $c_v = 0.05$ moves it to $\bar{\rho}^* \approx 5.7$; driving the jittered detector at the deterministic ridge forfeits about 55% of the information. Curves show the exact zero-jitter law or the long-window jitter law; markers show Monte Carlo.

for every finite T and every iid hidden hold law H independent of λ .

Equation (1) identifies the loss caused by count-only observation: uncertainty about how long the detector was active remains after conditioning on the recorded count. The deterministic detector confines this uncertainty to the terminal phase, whereas random holds accumulate it throughout the exposure. The identity requires amendment when holds depend on the load or scene, are observed by the electronics, are serially correlated, or when the counter is paralyzable.

Theorem 2 (Long-window information rate)

Under standard nonlattice renewal local-limit conditions,

$$\frac{I_N(\log \lambda; T)}{\nu} \rightarrow J_\infty(\rho, c_v), \quad (2)$$

$$J_\infty(\rho, c_v) = \frac{\rho}{(1 + \rho)(1 + c_v^2 \rho^2)}, \quad \rho = \lambda \tau.$$

The limiting loss depends on H through its mean and variance and remains extensive as $\nu \rightarrow \infty$ whenever $c_v > 0$.

The supplement proves both statements and gives the regularity conditions. In particular, the long-window result is intended for regular finite-variance iid hold families; it is not asserted for heavy-tailed, drifting, correlated, observed, or load-dependent recovery times.

3 Jitter-capped optimal load

Theorem 3 (Exact jitter-limited optimum) For every fixed $c_v > 0$, $J_\infty(\rho, c_v)$ has a unique interior maximum ρ_c satisfying

$$c_v^2 \rho_c^2 (1 + 2\rho_c) = 1, \quad (3)$$

with small-jitter expansion

$$\rho_c = 2^{-1/3} c_v^{-2/3} - \frac{1}{6} + \frac{2^{1/3}}{36} c_v^{2/3} + O(c_v^{4/3}). \quad (4)$$

Thus zero hold-time variance is the critical manifold on which the optimal load continues to grow with exposure length; every fixed nonzero hidden variance yields a finite long-window optimum. The exponent pair $(1/3, -2/3)$ applies within the regular finite-variance iid hidden-hold class, not to the excluded detector models listed above.

3.1 Finite-window crossover

Matching the random-hold loss against the deterministic terminal-boundary loss in the joint regime $\nu \rightarrow \infty$, $c_v \rightarrow 0$, and $c_v^2 \nu = O(1)$ gives

$$\rho_*(\nu, c_v) \sim (6\nu)^{1/3} (1 + 12\nu c_v^2)^{-1/3}. \quad (5)$$

The deterministic limit recovers the cube-root ridge and the jitter-dominated limit recovers $2^{-1/3} c_v^{-2/3}$. Equation (5) is a matched-asymptotic prediction, not a universal finite-window theorem; the supplement states the uniform expansion under which it follows.

3.2 Numerical verification

Predictions were registered before the refined simulations. At $\nu = 2000$, fresh points at $c_v = 0.02$ and 0.2 agreed with their predicted peaks to 3% and 6%, respectively. A continuous-peak run and an independent replication then gave a pooled log-log slope of -0.658 , within 1.3% of $-2/3$ (Fig. 2). Full grids, run sizes, the two-run peak table, and the recorded refutation of an intermediate multiplier hypothesis are reported in the supplement.

4 Global power-control policy

For a detector with measured recovery law H , define the operating target as the smallest maximizer of its finite-window count-information kernel over the admissible load interval,

$$\rho_R(\nu, H) = \min \arg \max_{\rho \in \mathcal{R}_{\text{adm}}(\nu)} J_H(\rho, \nu). \quad (6)$$

A common low-cost pre-scan estimates mean bucket brightness. One global source multiplier per dwell then places the mean load at ρ_R , followed by a single global safety clip; there is no per-pattern servo. The imaging campaign uses the deterministic recovery kernel to isolate this operating-point effect, for which $\rho_R(2000) = 22.25$. In a detector with nonzero measured jitter, the same policy replaces that target with the jitter-capped maximizer.

The resource interpretation is essential. At fixed dwell, ridge tracking is a power-for-time policy: it deliberately increases incident dose to obtain more information before the acquisition deadline. Under a hard incident-photon budget, by contrast, the relevant scalar efficiency is information per incident load, which favors lower operating points. The campaign therefore reports incident dose and detected counts beside every quality or speed benefit and makes no equal-photon-budget claim.

Illumination geometry is held fixed in the paired operating-point tests. A separate local design-measure analysis asks whether scene-adapted patterns can improve a log-determinant information surrogate under dose and conditioning constraints. Its generalized KKT system, dictionaries, guards, rounding, and solver checks are given in the supplement; because that analysis is descriptive, it is used only to expose remaining geometry headroom.

5 Preregistered imaging campaign

The campaign uses 24 fresh simulated scenes from six structural families, five paired measurement seeds, and nine dwell budgets $\nu \in \{5, 10, 20, 50, 100, 200, 500, 1000, 2000\}$. Every operating-point arm uses the same balanced SCAT32 geometry, the same common pre-scan, the same main pattern multiset, and the same RQL reconstruction. The safe reference operates at mean load 0.05, the fixed-dwell comparator at 0.60, and the ridge arm uses the dwell-dependent global multiplier of Eq. (6). Complete exposure accounting, scene generation, context arms, and immutable manifests are in the supplement.

The fixed-dwell primary compares ridge control with the 0.60 comparator at $\nu = 2000$. It passes when the median paired radiometric-PSNR gain is at least 1 dB, its family-stratified bootstrap lower bound exceeds zero, and at least 18 of 24 scenes are positive; numerical or guard failures remain in the cohort as nonpositive. The elapsed-time secondary compares ridge control with the 0.05 safe reference using frozen equal-weight isotonic fits, a Q90 target, and a pre-declared censoring taxonomy. It passes when median $S_{\text{gate}} \geq 3$, the bootstrap lower bound exceeds one, and at least 18 of 24 scenes have $S_{\text{gate}} > 1$. The exact endpoint names and full analysis specification are preserved in the supplement; the operating-point endpoint is `RIDGE_OPERATING_PASS`.

The separate geometry analysis assigns each of 480 full-conditioning cells one of three descriptive statuses: near-optimality certified, a feasible design with im-

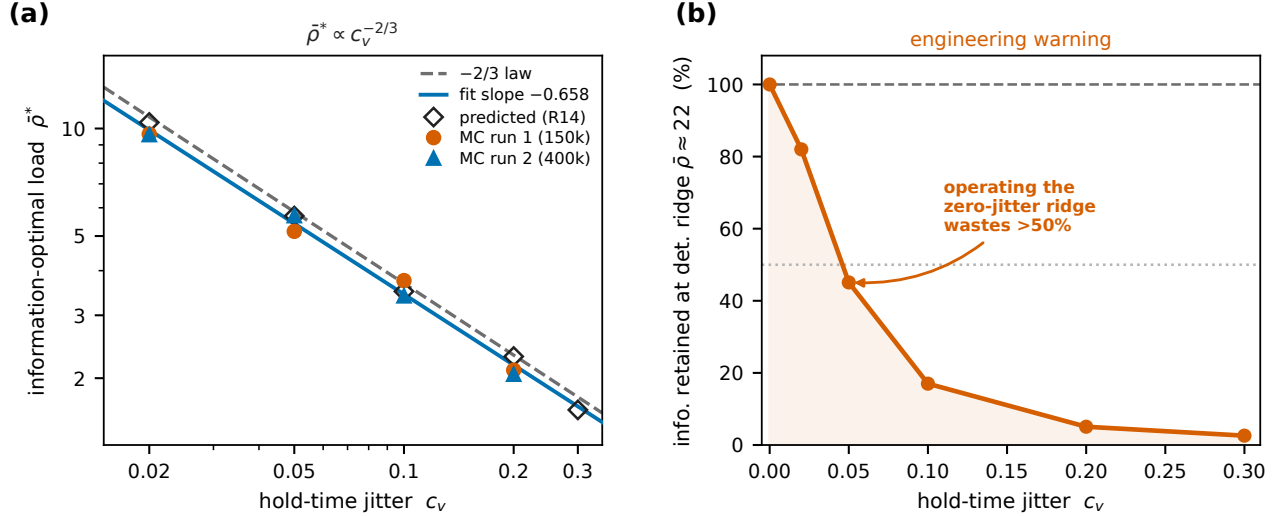


Figure 2: **Numerical verification of the jitter cap.** (a) Information-optimal load $\bar{\rho}^*$ versus hold-time jitter c_v at $\nu = 2000$. Two independent Monte-Carlo runs and the preregistered matched-asymptotic prediction follow a pooled fitted slope -0.658 , consistent with the theorem’s $-2/3$ power law. Marker and line styles distinguish theorem, prediction, and simulation. (b) Fraction of count information retained when a jittered detector is operated at the zero-jitter ridge $\bar{\rho} \approx 22$. Retention falls below one-half by $c_v = 0.05$ and to a few per cent by $c_v = 0.2$, showing why the operating load must track the detector’s own recovery-time variation.

provement above the frozen bar found, or numerically unresolved. It has no PASS/FAIL endpoint. The status definitions, full per-cell fields, compute protocol, and development branch are reported in the supplement.

5.1 Analysis correction and provenance

After the campaign was unblinded, an independent peer audit found that the frozen scoring implementation did not reproduce three parts of the preregistered analysis. It placed the Q90 crossings on $\log(\nu\rho)$ rather than on the frozen elapsed optical time $\log T_{\text{opt}}$; it resampled family labels rather than retaining all six fixed strata with paired within-image seed resampling; and it used a nonstandard isotonic implementation while omitting the frozen censoring taxonomy. The immutable tag, raw cells, and original analyzer outputs were preserved unchanged. A dated correction artifact recomputed the endpoints from the same frozen raw cells using the preregistered axis and machinery; the promoted independent implementation and an in-project rerun agreed on all 18 audited numerical outputs with no deep differences. Scientific claims and tables use the spec-conformant corrected results, while the original outputs remain available in the provenance record. The source-of-record bundle is `results/round63_m1/CORRECTION_2026-07-19/`, specifically `M1_VERDICTS_SPEC_CORRECTED_R19.json` and `ANALYSIS_CORRECTION_DISCLOSURE.md`.

6 Imaging results

6.1 Fixed-dwell quality and elapsed-time speed

The preregistered operating-point endpoint passed: at $\nu = 2000$, ridge control improved median fixed-dwell radiometric quality over the matched 0.60 comparator by $+1.87$ dB, with a family-stratified lower bound of 1.41 dB and 19 of 24 scenes positive. The same operating point used $37.1\times$ the incident dose and $2.6\times$ the detected counts. Using the preregistered elapsed optical-time coordinate, $T_{\text{opt}} = M_{\text{total}}\nu\tau$, and the frozen Q90/PAVA/censoring procedure, the ridge time-to-quality endpoint passed: the median conservative speed ratio was $19.13\times$, the family-stratified nested-bootstrap 2.5th-percentile bound was $18.33\times$, and 21 of 24 scenes had $S_{\text{gate}} > 1$. These are fixed-dwell and elapsed-time benefits bought with power, not photon-efficiency gains.

(The post-unblinding implementation-to-specification correction and its provenance are reported in the preceding “Analysis correction and provenance” subsection.)

The fixed-dwell response is scene dependent. The median family gain ranges from $+6.37$ dB for maze-like scenes to -4.18 dB for contour scenes, and the per-scene gain correlates with measured bucket contrast at $+0.64$. No load, ceiling, or reconstruction guard explains the negative cases; they instead delimit where one global

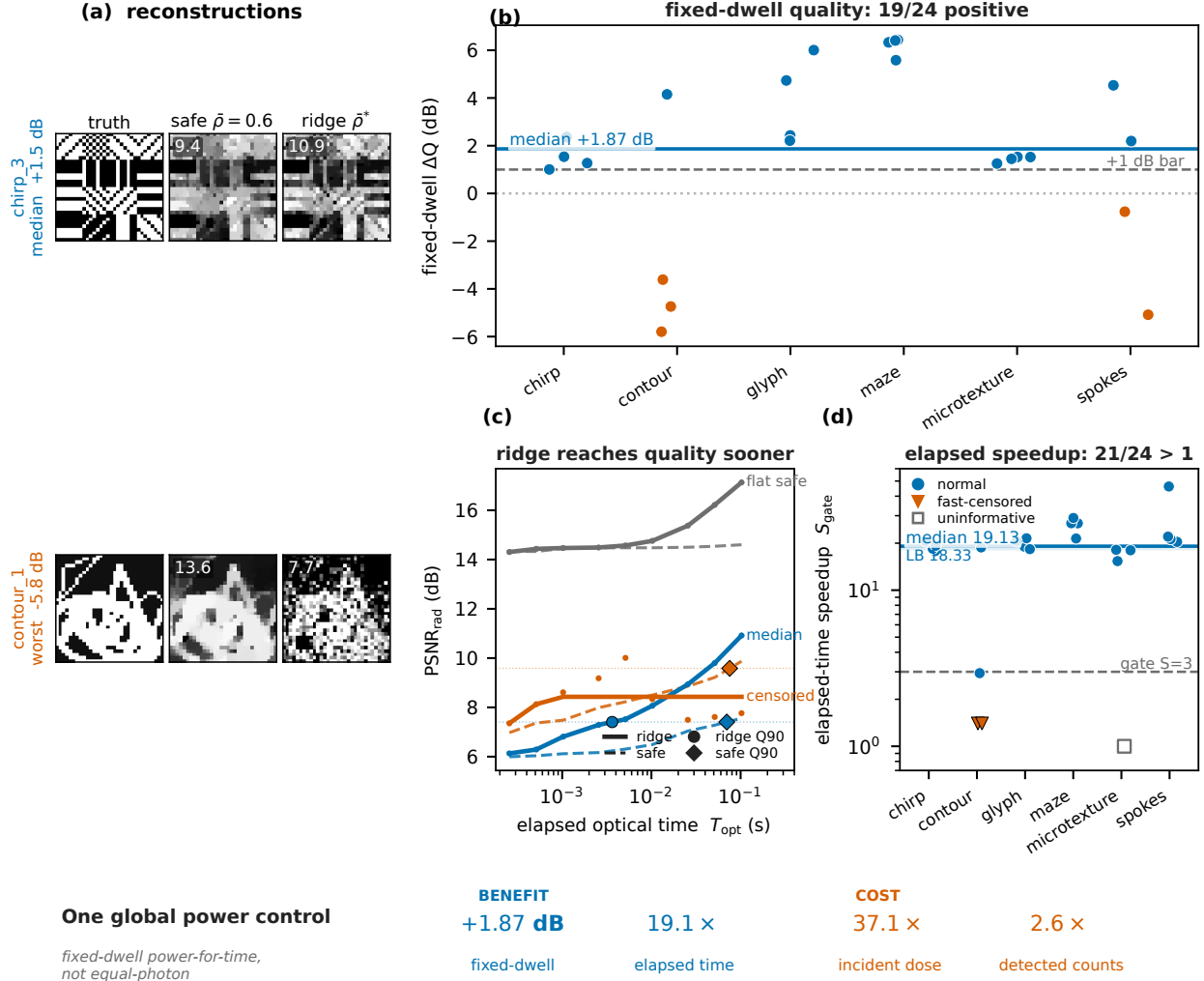


Figure 3: **One global power control trades incident dose for elapsed-time speed and scene-dependent fixed-dwell quality.** (a) Truth, safe-comparator, and ridge reconstructions for a median and a worst fixed-dwell scene, with radiometric PSNR on each crop. (b) Per-scene fixed-dwell gain ΔQ at $\nu = 2000$: the campaign median is +1.87 dB and 19 of 24 scenes are positive; negative cases remain visible. (c) Quality versus elapsed optical time T_{opt} for a median-speed scene and two boundary cases. (d) Elapsed-time speedup S_{gate} for all 24 scenes: median 19.13 and family-stratified lower bound 18.33, with 21 of 24 above one. The benefits use 37.1 $\times$ incident dose and 2.6 $\times$ detected counts; this is a power-for-time comparison, not an equal-photon-budget comparison.

Table 1: Benefit and resource cost at the terminal dwell. ΔQ compares ridge control with the matched 0.60 fixed-dwell comparator; S_{gate} compares it with the safe reference. Dose and detected counts are normalized to the 0.60 comparator.

Policy	achieved $\bar{\rho}$	incident dose	detected counts	ΔQ (dB)	S_{gate}
Safe reference	0.05	0.08 $\times$	0.13 $\times$	—	1.00
Fixed-dwell comparator	0.60	1.00 $\times$	1.00 $\times$	0.00	—
Global ridge control	22.25	37.1 $\times$	2.6 $\times$	+1.87	19.13

multiplier is insufficiently aligned with scene structure. The complete per-family and per-scene tables are in the supplement. A post-hoc incident-exposure version of the Q90 analysis gave a median ratio 0.279 and no

verdict, confirming that the elapsed-time speedup is not an incident-photon saving.

7 Geometry-design headroom

Scene-adapted geometry retained measurable headroom in two deliberately separate searches. A development-only dose-uniform construction improved the local determinant over balanced SCAT32 in all 12 family-anchor cells, giving D-efficiency lower bounds from $1.64\times$ to $2.51\times$. Full-conditioning feasible searches on fresh campaign cells also found concrete designs with larger local information. These are existence results for a local information surrogate, not confirmatory image endpoints or estimates of a population optimum.

The full-conditioning analysis remained descriptive: 0 of 480 cells certified SCAT32 near-optimality, 299 produced feasible counterexamples, and 181 were numerically unresolved. It therefore establishes neither SCAT32 near-optimality nor a reliable adaptive replacement. The retired adaptive path, development replication table, status distribution, conic construction, solver protocol, and all caveats appear in the supplement.

8 Discussion and limitations

The central practical result is a detector-specific operating rule. Hidden hold-time variation changes the information optimum from a dwell-dependent deterministic ridge to a finite jitter cap, and a single global power control can track that optimum without per-pattern feedback. In the paired simulations, the rule delivered a large elapsed-time benefit and a smaller, strongly scene-dependent fixed-dwell benefit. Their resource cost is equally central: the policy converts available incident power into shorter dwell and is not a photon-budget advantage.

Geometry adaptation remains an open second axis. Feasible designs demonstrate that balanced illumination is not known to exhaust the available local information, but the descriptive analysis does not identify a stable adaptive policy or validate image-level gains. A future method must jointly control geometry, conditioning, and the jitter-limited scalar operating point, then be tested on fresh scenes and hardware.

8.1 Limitations

First, the terminal ridge policy requires substantial peak and average optical power: the reported fixed-dwell operating point uses $37.1\times$ the incident dose of its comparator, so source limits, spatial-light-modulator

limits, laser safety, or sample damage may bind before the detector optimum is reached. Second, the load target is only as reliable as the recovery-time calibration; the theory assumes a stable iid hidden-hold law, whereas drift, correlation, paralyzable behavior, or load-dependent recovery requires a different kernel. Third, all imaging results are simulation-only. Exact renewal simulation, paired seeds, and preregistration do not replace optical-bench validation.

8.2 Immediate bench test

The count-information law can be tested before a complete imaging system is built. Time-tagged recovery intervals estimate c_v ; repeated fixed-dwell frames acquired during a source-power sweep then trace $J(\rho)$. For $c_v = 0.05$ and $\nu = 2000$, the predicted peak is near $\rho = 5.7$ rather than 22, with roughly half the information lost at the deterministic ridge. A subsequent SPI experiment can compare the same balanced pattern sequence under safe and detector-specific global power control while recording incident dose, detected counts, and elapsed time.

Code and data availability

All simulation code, preregistration documents, manifests, correction records, and evidence bundles are available at [\[repo URL wording — user decision\]](#).

Acknowledgments

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